

Regularization

↳ Another method to overcome overfitting in Neural network

Why overfitting happens?

↳ The complexity of my model/Neural network

Ways to solve overfitting

- Adding more data
- Reducing complexity of model

→ Add more rows

→ Data Augmentation

↳ modify current data to make on inc data

→ Dropout

→ Early Stopping

→ Regularization

→ L_1

→ L_2 → mostly used

→ $L_1 + L_2$

We know, we are when we talk about ANN

we used to get weight & bias

by minimizing loss function

L → mse

↳ Binary cross entropy

C_{oft}

$$C = \frac{1}{n} \sum_{i=1}^n L(y_i, \hat{y}_i)$$

cost function

This is regular CF

Just in L_2 Regularization we add Penalty term in CF

$$C = L + \text{Penalty term}$$

Same weight square sum
but not of bias

$$C = L + \frac{\lambda}{2n} \sum_{i=1}^K ||w_i||^2$$

$\lambda \rightarrow$ hyperparameters

$\lambda = 0$
The Same
CF, No Regular

The more λ the
weightage of Penalty term

inc more
the model will
underfit

$$C = \sum L(y_i - \hat{y}_i) + P$$

weights value towards $\rightarrow 0$

for L_1

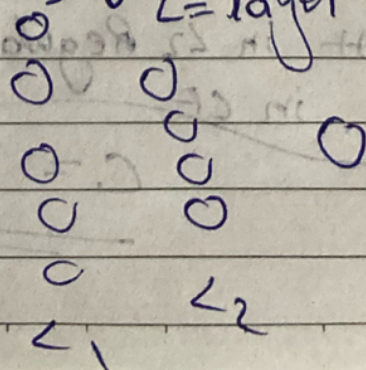
$$C = L + \frac{\lambda}{2n} \sum_{i=1}^K ||w_i||$$

$\rightarrow$ No Square

$$C = \sum_{i=1}^n L(y_i - \hat{y}_i)^2 + \sum_{d=1}^L \sum_{i=1}^n \sum_{j=1}^n ||w_{ij}^{(d)}||^2$$

more accurate

$i = \text{row}$
 $j = \text{col}$
 $L = \text{layer}$



Intuition behind Regularization

$$w_n = w_0 - \eta \frac{dL}{dw_0} \rightarrow \text{G.D}$$

$$L' = L + \frac{\lambda}{2} w_0^2 \leq 1 \text{ will}^2$$

↓ for single training example

$$\begin{aligned} \frac{dL'}{dw_0} &= \frac{dL}{dw_0} + \frac{\lambda}{2} 2w_0 \\ &= \frac{dL}{dw_0} + \lambda w_0 \end{aligned}$$

$$w_n = w_0 - \eta \left(\lambda w_0 + \frac{dL}{dw_0} \right)$$

$$w_n = w_0 - \eta \lambda w_0 - \eta \frac{dL}{dw_0}$$

$$w_n = w_0 - \eta \frac{dL}{dw_0} = (1 - \eta \lambda) w_0 - \eta \frac{dL}{dw_0}$$

$$\boxed{1 - \eta \lambda} \rightarrow \text{weight decay}$$

→ ~~Hint~~
It is +ve

So new w_0 will be less than old w_0

$$w_n < w_0$$

Weight decrease in every epoch towards zero, but never be zero